Report from Inspect Phase of Data Cleaning Exercise

Column Names

S/N	Checklist	Steps	Outcomes
1	***Row & Column Basics***	Total rows and columns	Columns = 25, Rows = 2823
		What one row represents, likely one order line item, not one order.	Rows represent order line item.
		Confirm '**Order Number**' repeats across multiple line items	Yes
2	***Data Types***	Which Columns are text vs number vs date	whole_No = 7, text = 13, decimal_No = 2, date & time = 1, any = 2
		Check '**Order Date**', is itactually stored as date, or as text?	**Order Date** is stored in "Date & Time" data format
		QTR ID', '**Month ID**', '**Year ID**' - Are they numbers or text?	Saved as "whole_No"
3	***Missing / Blank Values***	Count blanks per column	**Address Line 2** - 2521/2823 (89%) **State** - 1486/2823 (53%) **Postal Code** - 76/2823 (3%)
		Especially '**Address Line 2**', '**State**', '**Postal Code**', '**Territory**', '**Deal Size**'	Found blank (NULL) values in "**Address line 2, State, Postal Code**" columns
		Is '**Address Line 2**' mostly blank? If so, likely drop or flag	Yes mostly blank
4	***Duplicates***	Full-row duplicates	Duplicates found (Just different items in the same Order Number), Not true duplicates
		Duplicate '**Order Number**' + '**Order Line Number**' combinations - Those would be the true duplicates	None Found
		Repeated '**Order Number**' alone is expected, not an error	True (Found duplicate when i used **Order Number** as the criteria)
5	***Categorical Consistency***	**Territory**' and '**Country**' casing/spelling - e.g USA vs US, EMEA vs Emea	None Found
		Status' unique values - Shipped, Cancelled, On Hold, Disputed, In Process, e.t.c	Yes unique values found in **Status** (Disputed, In Process, On Hold, Shipped, Resolved, Cancelled)
		"**Deal Size**" values - Small, Medium, Large only?	Yes
		"**Product Line**" - any near-duplicates or trailing spaces?	None Found (but to be safe, i will apply TRIM to all the columns)
6	***Numerical ranges & logic***	"**Quantity Order**" - any zero, negative, or extreme values?	None Found
		"**Price Each**" - any zero or negative?	None Found
		"**Sales**" - does it roughly equal "**Quantity Ordered**" * "**Price Each**"?	I noticed a pattern: when **Price Each** is 100, the **Sales** value is incorrect, and I need to discuss it with stakeholders to get answers. e.g **Quantity Ordered**(49)*__Price Each__(100) should be (4,900), but the **Sales** value is (5,205.27).
		"**MSRP**" - any value below "**Price Each**" missing?	None Found
		"**Order Line Number**" - does it start at 1 per order and increment sensibly?	Yes (1-18)
7	***Date range & seasonality***	Min and Max "**Order Date**"	5/31/2005 - Newest Date 1/6/2003 - Oldest Date
		Do "**Year ID**", "**QTR ID**", and "**Month ID**" align with "**Order Date**" ?	Yes
		Any dates outside expected range?	None Found
8	***Relationship Checks***	Does each "**Product Code**" Map to one "**Product line**" and one "**MSRP**"?	Yes
		Does each "**Customer Name**" map to one "**Territory** / **City** / **State**" consistently?	Yes
		Any "**Order Number**" with missing "**Customer Name** or **Sales**"?	None Found